Special Summer Integrated Learning Programme

Mathematics

Exponential and Logarithmic Functions

Week 2 Day 2 AM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

Extending exponentials to $\mathbb{R}$

Although we have defined exponential a^x last week, but it's only defined for rational exponents x. It's not defined for irrational x. For instance, what is the meaning of 3^π or $2^{\sqrt{2}}$? We are tempted to say 3^π means 3 multiplied by itself π times, but this statement doesn't make any sense! (How many times is π times?)

Thus, we need to rigorously extend the exponential to all real exponents. There are many equivalent ways to do this. Here, we'll define it using the most intuitive idea. Suppose we want to define a^π for some $a > 0$. Since $\pi = 3.14159265358\ldots$, we can successively approximate a^π by the following rational exponents:

$$a^3, a^{3.1}, a^{3.14}, a^{3.141}, a^{3.1415}, \ldots$$

It can be shown by using advanced mathematics that there is exactly one real number that these exponentials approach. We define a^π to be this number.

Exponential functions

Thus, in general for any $a > 0$ and any real exponent x, we'll define a^x as the unique real number approached by the approximation similar to the example given in the previous slide.

Definition:

The ***exponential function with base*** a is the function

$$f : \mathbb{R} \to (0, \infty), \;\; f(x) = a^x,$$

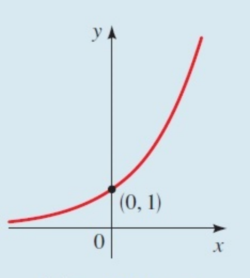
where $a > 0$ and $a \neq 1$ (if $a = 1$, $a^x = 1$ is just the constant function equals 1). It can be shown that the laws of exponents introduced last week are still true when the exponents are real.

Fun facts:

- It's possible to extend the exponentials to negative a, but this requires complex numbers. For intance, $(-1)^{\frac{1}{2}} = i$.
- It's possible to extend the exponentials to complex non-real a and x. For instance $i^i = e^{-\frac{\pi}{2}}$.

Graphs of exponential functions $a^x, a > 1$

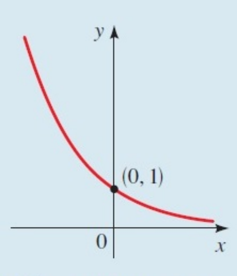
If $a > 1$, then $f(x) = a^x$ gets arbitrarily large when $x \to \infty$, as we are multiplying $a > 1$ a large number of times. When $x \to -\infty$, the negative exponent means division, so $f(x)$ approaches 0 as we are dividing an arbitrarily large number. So $y = 0$ is a horizontal asymptote of $f(x) = a^x$.



$f(x) = a^x$ for $a > 1$

Graphs of exponential functions $a^x, a < 1$

If $0 < a < 1$, then $f(x) = a^x$ approaches 0 when $x \to \infty$, as we are multiplying $0 < a < 1$ a large number of times. When $x \to -\infty$, the negative exponent means division, so $f(x)$ gets arbitrarily large, as we are dividing a small positive number. Again, $y = 0$ is a horizontal asymptote of $f(x) = a^x$.



$f(x) = a^x$ for $0 < a < 1$

Exponential growth

Since $f(x) = a^x \to \infty$ as $x \to \infty$ for $a > 1$, how fast does a^x increase compared to polynomials? You have most likely heard people describing rapid increases as exponential growths, and indeed a^x increases faster than any polynomials as $x \to \infty$. This result can be proven using limits and differential calculus.

Theorem

For any $a > 1$ and polynomial $P(x)$ of any degree with a positive leading coefficient,

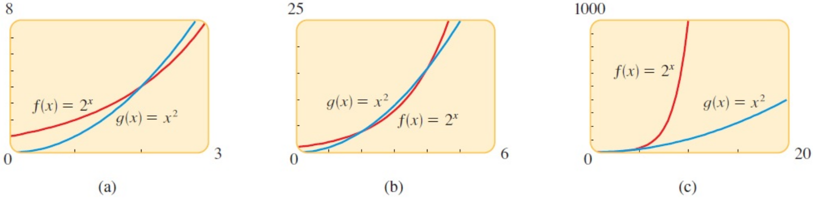
$$a^x - P(x) \to \infty \quad \text{and} \quad \frac{a^x}{P(x)} \to \infty \quad \text{as} \quad x \to \infty,$$

or equivalently

$$\frac{P(x)}{a^x} \to 0 \quad \text{as} \quad x \to \infty.$$

Exponential growth

Note that the previous theorem says that the values of exponentials dominate polynomials as $x \to \infty$, but it doesn't say anything about the case when x is not large. In fact, the value of an exponential function can be smaller than the value of a polynomial at the same x.



Fun fact: Exponential growths are applied in compounding interest. Albert Einstein (unverified) once said "Compound interest is the eighth wonder of the world. He who understands it, earns it; he who doesn't, pays it".

Compound interest

If we invest a principal P at an interest rate i per time period, then after one time period the interest is Pi, and the total amount A_1 of money is

$$A_1 = P + Pi = P(1+i).$$

If the interest is reinvested, then the new principal is $P(1+i)$, and the amount after another time period is

$$A_2 = P(1+i) + P(1+i)i = P(1+i)^2.$$

Similarly, after the third time period the amount is

$$A_3 = P(1+i)^3.$$

In general, after k periods the amount is

$$A_k = P(1+i)^k.$$

Compound interest

For most bank deposit accounts, they offer an annual interest rate r per year but they compute the interest monthly. The natural way to split the interest to each month is to divide r by 12, i.e. $i = \frac{r}{12}$ as the interest rate per month. By using the formula from the previous slide, the total amount of money after t years will be

$$A(t) = P\left(1 + \frac{r}{12}\right)^{12t},$$

as the total number of times the interest is compounded in t years is $12t$ (12 times each year).

Activity 1

A bank offers an annual interest of 3.5% for its deposit account. With a principal of $\$10000$, find the total amount of money in the account after 5 years, if

(a) the interest is compounded annually,

(b) the interest is compounded monthly.

Activity 1: solution

We are given $P = 10^4$, $r = 0.035$ and $t = 5$.

(a) If the interest is compounded annually, then $i = 0.035$ and the interest is compounded 5 times after 5 years, so the total amount in the account is

$$10^4(1 + 0.035)^5 \approx 11876.86.$$

(b) If the interest is compounded monthly, then $i = \frac{0.035}{12}$ and the interest is compounded 5×12 times after 5 years, so the total amount in the account is

$$10^4 \left(1 + \frac{0.035}{12}\right)^{5\times 12} \approx 11909.43.$$

Compound interest

As we see in the previous activity, compounding the interest monthly will give us a higher return compared to compounding the interest annually. You might start to become greedy and request the bank to compute the interest more frequently. Let's see whether this will make you infinitely rich.

If the interest is compounded n times per year with an annual interest of r, then the interest is $\frac{r}{n}$ each time it's calculated. After t years, the interest is compounded nt times, so the total amount of money is

$$A(t) = P\left(1 + \frac{r}{n}\right)^{nt}.$$

Activity 2

With the same setting as the previous activity, i.e. $r = 3.5\%$, $P = \$10000$, and $t = 5$ years, find the total amount of money in the account after 5 years if

(a) the interest is compounded daily, i.e. $n = 365$.

(b) the interest is compounded hourly, i.e. $n = 365 \times 24 = 8760$.

(c) the interest is compounded every minute, i.e. $n = 365 \times 24 \times 60 = 525600$.

(d) the interest is compounded every second, i.e. $n = 365 \times 24 \times 60 \times 60 = 31536000$.

What do you observe? Does the money increase without bound?

Activity 2: solution

(a)
$$A(t) = 10^4\left(1+\frac{0.035}{365}\right)^{365\times 5} \approx 11912.3622.$$

(b)
$$A(t) = 10^4\left(1+\frac{0.035}{8760}\right)^{8760\times 5} \approx 11912.4580.$$

(c)
$$A(t) = 10^4\left(1+\frac{0.035}{525600}\right)^{525600\times 5} \approx 11912.4621.$$

(d)
$$A(t) = 10^4\left(1+\frac{0.035}{31536000}\right)^{31536000\times 5} \approx 11912.4622.$$

The amount of money doesn't seem to increase without bound but to approach certain value. We'll show later that it indeed will approach the value

$$10^4 e^{0.035\times 5} = 11912.46216612358\ldots.$$

e is a constant called Euler's number, which we'll defined in the next section.

Outline

The Euler's number e

As we see from the previous activity, the compound interest approaches certain value when it's computed more and more frequently. In fact, if we start with a principal $P = 1$ and an annual interest $r = 1$, the money we get after one year as we compound the interest n times per year is given by

$$\left(1 + \frac{1}{n}\right)^n.$$

n	$\left(1 + \frac{1}{n}\right)^n$
1	2.00000
5	2.48832
10	2.59374
100	2.70481
1000	2.71692
10,000	2.71815
100,000	2.71827
1,000,000	2.71828

This value approaches a constant $2.718281828459045\,23536\ldots$ as $n \to \infty$.

The natural exponential function

Definition:

The number approached by $\left(1+\frac{1}{n}\right)^n$ as $n \to \infty$ is called ***Euler's number*** and it's denoted as e.

$$e = 2.718281828459045235 36\ldots$$

The ***natural exponential function*** is the exponential function

$$f(x) = e^x$$

with base e.

⚠ Do not call e Euler's constant. Euler's constant γ is a different mathematical constant defined in advanced mathematics.

Fun fact: Mathematicians define $f(x) = e^x$ in many different ways, but they are all equivalent. For instance, it's the unique function whose derivative is itself and equals 1 at $x = 0$.

Continuously compounded interest

Recall that if we compound the interest n times per year, the money we get after t years is

$$A(t) = P\left(1+\frac{r}{n}\right)^{nt},$$

where P is the principle and r is the annual interest. We rearrange some terms and denote $\frac{n}{r}$ as m:

$$A(t) = P\left(1+\frac{r}{n}\right)^{nt} = P\left(\left(1+\frac{1}{\frac{n}{r}}\right)^{\frac{n}{r}}\right)^{rt} = P\left(\left(1+\frac{1}{m}\right)^{m}\right)^{rt}.$$

Since by definition, $\left(1+\frac{1}{m}\right)^{m}$ approaches e as $m \to \infty$, so if we compound the interest continuously, i.e. $n \to \infty$, then $\frac{n}{r} = m \to \infty$ also and thus $A(t)$ approaches Pe^{rt}.

Continuously compounded interest

Definition:

Continuously compounded interest is calculated by the formula

$$A(t) = Pe^{rt},$$

where

$$\begin{aligned} A(t) &= \text{total amount after } t \text{ years}, \\ P &= \text{principal}, \\ r &= \text{annual interest rate}, \\ t &= \text{number of years}. \end{aligned}$$

E.g.: If we compound the interest in activity 2 continuously, then $A(5) = 10^4 e^{0.035\times 5} = 11912.46216612358\ldots$ as we claimed before.

Activity 3

Animal populations are not capable of unrestricted growth because of limited habitat and food supplies. Under such conditions the population $P(t)$ follows a ***logistic growth model***:

$$P(t) = \frac{d}{1 + ke^{-ct}},$$

where c, d, and k are positive constants. For a certain fish population in a small pond $d = 1200$, $k = 11$, $c = 0.2$, and t is measured in years. The fish were introduced into the pond at time $t = 0$.

(a) How many fish were originally put in the pond?

(b) Find the population after $10, 20$, and 30 years.

(c) Evaluate $P(t)$ for large values of t. What value does the population approach as $t \to \infty$?
Hint: recall the asymptotic behavior in slides 5 and 6.

Activity 3: solution

(a) Since the fish were introduced at $t = 0$, the original number of fish is

$$P(0) = \frac{d}{1 + ke^0} = \frac{d}{1+k} = \frac{1200}{1+11} = 100.$$

(b)

$$P(10) = \frac{d}{1 + ke^{-10c}} = \frac{1200}{1 + 11e^{-2}} \approx 482.$$

$$P(20) = \frac{d}{1 + ke^{-20c}} = \frac{1200}{1 + 11e^{-4}} \approx 999.$$

$$P(30) = \frac{d}{1 + ke^{-30c}} = \frac{1200}{1 + 11e^{-6}} \approx 1168.$$

(c) Since $e > 1$, so e^{-ct} approaches 0 as $t \to \infty$ (because the exponent $-ct \to -\infty$). As a result,

$$P(t) \text{ approaches } \frac{d}{1 + k \cdot 0} = d = 1200 \quad \text{as} \quad t \to \infty.$$

Outline

Inverse of exponential functions

Since exponential functions $f : \mathbb{R} \to (0, \infty)$, $f(x) = a^x$ are one-to-one (increasing for $a > 1$, decreasing for $a < 1$) and onto, thus they are always invertible. The inverse function of an exponential function is called the logarithmic function.

Definition:

Let $a > 0$ and $a \neq 1$. The ***logarithmic function with base*** a, denoted by $\log_a : (0, \infty) \to \mathbb{R}$, is defined as the inverse function of $f : \mathbb{R} \to (0, \infty)$, $f(x) = a^x$, i.e.

$$y = \log_a(x) \Leftrightarrow a^y = x.$$

In layman term, $\log_a(x)$ is the exponent to which the base a must be raised to give x.

⚠ The domain of $\log_a(x)$ is $(0, \infty)$, i.e. it's only defined for $x > 0$.

Properties of logarithms

- Since $a^0 = 1$, hence
$$\log_a(1) = 0.$$
- Since $a^1 = a$, hence
$$\log_a(a) = 1.$$
- The defining property of inverse functions
$$\left(f \circ f^{-1}\right)(x) = x \quad \text{and} \quad \left(f^{-1} \circ f\right)(x) = x$$
for $\log_a(x)$ would be
$$a^{\log_a(x)} = x \quad \text{and} \quad \log_a(a^x) = x.$$

The common logarithm

Some logarithmic functions with specific bases are used frequently in science and mathematics, thus we give them specific notations.

Definition:

The logarithm with base 10 is called the ***common logarithm*** and is denoted by:

$$\lg x := \log_{10}(x).$$

$\lg$ is useful in telling us the magnitude of a number. If we write $\lg x = n + r$, where $n = \lfloor \lg x \rfloor$ (recall that $\lfloor \cdot \rfloor$ is the greatest integer function) and $0 \leq r < 1$ is the decimal part of $\lg x$, then

$$x = a \times 10^n, \text{ where } 1 \leq a = 10^r < 10.$$

In particular, if $\lg x > 0$, then one plus the integer part of $\lg x$ is the number of digits of the integer part of x (written in base 10).

E.g.: $\lg 1000 = 3$, $\lg 12345 = 4.091\ldots$, $\lg \frac{2024}{3} = 2.829\ldots$, $\lg 0.1 = -1$, $\lg \frac{1}{3} = -0.4771\ldots$, $\lg \frac{1}{17} = -1.2304\ldots$.

The natural logarithm

Definition:

The logarithm with base e is called the ***natural logarithm*** and is denoted by:

$$\ln x := \log_e(x).$$

Remark: $\ln$ is formally pronounced as "ell en" or natural log. Informally, it's pronounced as "lawn".

$\ln$ is useful in solving equations that involve the natural exponential function.

E.g.: To solve $25 = 24 + 76e^{-0.1t}$, we proceed as follows

$$\begin{aligned} 76e^{-0.1t} &= 25 - 24 \quad &\Leftrightarrow\quad e^{-0.1t} &= \frac{1}{76} \\ \Leftrightarrow \quad -0.1t &= \ln\left(\frac{1}{76}\right) \quad &\Leftrightarrow\quad t &= -10\ln\left(\frac{1}{76}\right) \\ & &\Leftrightarrow\quad t &= 43.3073\ldots \end{aligned}$$

Activity 4

(a) An investment offers 4% of annual interest. Suppose the interest is compounded continuously, find the time needed to double the principal.

(b) Find the time needed for the population of the fish in activity 3 to reach 888.

Activity 4: solution

(a) Let the principal be P, so the money after t years is $A(t) = Pe^{0.04t}$. To double the principal, we want $A(t) = 2P$, thus

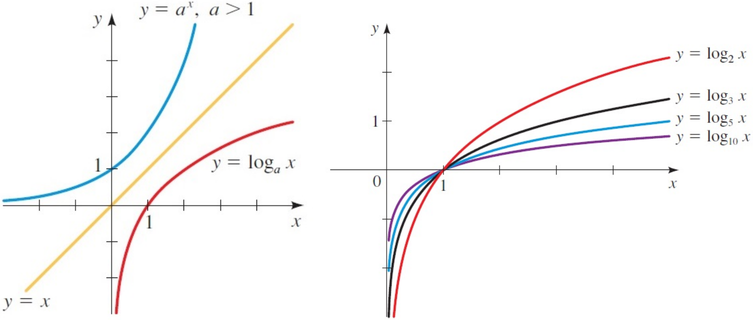
$$2P = Pe^{0.04t} \quad \Leftrightarrow \quad 2 = e^{0.04t}$$
$$\Leftrightarrow \quad \ln 2 = 0.04t \quad \Leftrightarrow \quad t = \frac{\ln 2}{0.04} \approx 17.33 \text{ years.}$$

(b) Since $P(t) = \frac{d}{1+ke^{-ct}}$ with $d = 1200, k = 11, c = 0.2$. For $P(t) = 888$,

$$888 = \frac{d}{1+ke^{-ct}} \quad \Leftrightarrow \quad 1 + ke^{-ct} = \frac{d}{888}$$
$$\Leftrightarrow \quad e^{-ct} = \frac{1}{k}\left(\frac{d}{888} - 1\right) \quad \Leftrightarrow \quad -ct = \ln\left(\frac{1}{k}\left(\frac{d}{888} - 1\right)\right)$$
$$\Leftrightarrow t = -\frac{1}{c}\ln\left(\frac{1}{k}\left(\frac{d}{888} - 1\right)\right) = -\frac{1}{0.2}\ln\left(\frac{1}{11}\left(\frac{1200}{888} - 1\right)\right) \approx 17.22 \text{ years.}$$

GRAPHS OF LOGARITHMIC FUNCTIONS

Since logarithmic functions are inverse functions of exponential functions, their graphs are simply the reflections of the graphs of exponential functions about the line $y = x$.



If $a > 1$, $\log_a(x) \to \infty$ as $x \to \infty$. On the other hand, $\log_a(x) \to -\infty$ as $x \to 0^+$, so $x = 0$ is a vertical asymptote of $\log_a(x)$.

Logarithmic growth

Since $f(x) = \log_a(x) \to \infty$ as $x \to \infty$ for $a > 1$, how fast does $\log_a(x)$ increase compared to polynomials? It can be shown using limits and differential calculus that $\log_a(x)$ increases slower than any non-constant polynomials with positive leading coefficients as $x \to \infty$.

Theorem

For any $a > 1$ and polynomial $P(x)$ of degree≥ 1 with a positive leading coefficient,

$$P(x) - \log_a(x) \to \infty \quad \text{and} \quad \frac{P(x)}{\log_a(x)} \to \infty \quad \text{as} \quad x \to \infty,$$

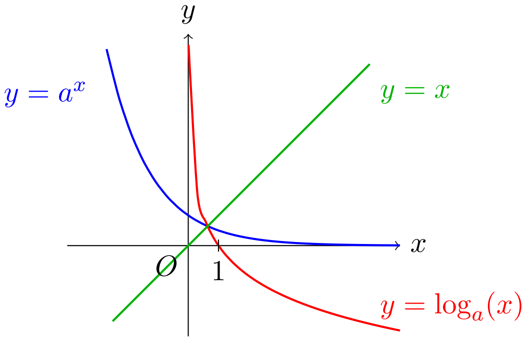
or equivalently

$$\frac{\log_a(x)}{P(x)} \to 0 \quad \text{as} \quad x \to \infty.$$

Activity 5

Sketch the graph of $f(x) = \log_a(x)$ for $0 < a < 1$ and describe the its behavior as $x \to \infty$ and $x \to 0^+$.

Activity 5: solution



For $0 < a < 1$, $\log_a(x) \to -\infty$ as $x \to \infty$. On the other hand, $\log_a(x) \to \infty$ as $x \to 0^+$, so $x = 0$ is a vertical asymptote of $\log_a(x)$.

Fun fact: Contrary to the case of $a > 1$, for $0 < a < 1$, the graphs $y = a^x$ and $y = \log_a(x)$ intersect at a point on the line $y = x$.

Outline

Summary

- Exponential Functions
  - extending exponentials to $\mathbb{R}$.
  - exponential functions and their graphs.
  - exponential growth.
  - compound interest.
- The Natural Exponential Function
  - the Euler's number e.
  - the natural exponential function.
  - continuously compounded interest.
- Logarithmic Functions
  - inverse of exponential functions.
  - Properties of logarithms.
  - the common logarithm and the natural logarithm.
  - graphs of logarithmic functions.
  - logarithmic growth.

Resources

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 4.1, 4.2, 4.3.